A059 — SST Quantum-Number Rosetta Mega Falsifier

v0.2.0 — SST Falsifier Framework v1.0.6

1 Identification and scope

A059 v0.2.0 is a self-contained, phased and scientifically nonblocking finite-model falsifier. It vendors framework v1.0.6, embeds E011 v0.3.0 STATIC_READY provenance, and adds canonical closed-braid dynamic probes. E011 remains a static source boundary; generated probes do not upgrade it to independent dynamics-ready evidence.

2 Research question

Can the registered knot/link topologies support reproducible dynamic operator candidates for family separation, signed central-core response, knot-specific tangent-map monodromy, circulation conjugation and a three-component same-sign axial-core container?

3 Competing hypotheses

The null is that apparent dynamic structure is construction-dependent, numerically unstable, universal rather than topology-specific, or insufficiently independent. The alternative requires at least one registered dynamic operator to survive its numerical/symmetry controls; integrated closure remains a separate stricter question.

4 Assumptions and boundary

All selected single-knot probes use the same closed-braid construction, smoothing and arclength normalization. The three-component candidate has one straight $+z$ core vortex through each ring centre, all three with equal positive circulation. The period lane uses only closed-loop line-integral/residue mathematics. No physical interpretation is assigned to that mathematical construction during blind analysis.

5 Symbols and units

All decision observables are dimensionless after the explicit v0.2.0 geometry/time normalization. The finite-core radius and arclength are expressed in the normalized probe coordinate system.

6 Registered equations

Machine-readable science contract

Research question Can self-contained canonical representatives of provenance-qualified E011 knot/link topologies support reproducible dynamic operators for family separation, signed external-core coupling, knot-specific tangent-map monodromy, circulation conjugation and a three-component same-sign core-vortex container, without semantic particle labels during blind analysis?

Objective Upgrade A059 from static/operator screening to finite-core dynamic probes while preserving nonblocking execution, E011's STATIC_READY boundary, and semantic reveal separation. The period lane uses only closed-loop line-integral/residue mathematics; no physical interpretation is assumed by that mathematical construction.

H_0 H0: the new dynamic observables are representation/construction artifacts, numerically unstable, universal kinematic identities, or insufficiently independent to support a multi-operator taxonomy.

H_1 H1: at least one preregistered dynamic operator survives its numerical and symmetry controls, while integrated closure is claimed only if every required independent operator sector closes.

Registered equations

E1: desingularized filament velocity

$$\mathbf{v}(\mathbf{x}) = \frac{\Gamma}{4\pi} \oint_K \frac{d\boldsymbol{\ell} \times (\mathbf{x} - \mathbf{X})}{(|\mathbf{x} - \mathbf{X}|^2 + a^2)^{3/2}} \tag{1}$$

Dimensional check: normalized finite-core Biot-Savart operator.

E2: reduced dynamic Jacobian

$$(J_K)_{ij} = \langle B_i,\, D\mathbf{v}_K[B_j] \rangle \tag{2}$$

Dimensional check: basis is normalized; all compared curves have equal total arclength.

E3: dynamic family fingerprint

$$F_K = (\log(1 + ||J_K||_F),\, \mathrm{rms}(\mathrm{Im}\,\lambda(J_K)/||J_K||_F)) \tag{3}$$

Dimensional check: dimensionless.

E4: central-axis period

$$N_C = \frac{1}{\Gamma_0} \oint_C \mathbf{v}_{\mathrm{core}} \cdot d\boldsymbol{\ell} \tag{4}$$

Dimensional check: circulation divided by circulation.

E5: dynamic external-core response

$$q_{\mathrm{dyn}} = \frac{\int_K \mathbf{v}_{\mathrm{self}} \cdot \mathbf{v}_{\mathrm{core}}\, ds}{\sqrt{\int_K |\mathbf{v}_{\mathrm{self}}|^2 ds \int_K |\mathbf{v}_{\mathrm{core}}|^2 ds}} \tag{5}$$

Dimensional check: normalized correlation; dimensionless.

E6: circulation-conjugation covariance

$$J_K(-\Gamma) = -J_K(+\Gamma), \quad \mathbf{v}(-\Gamma) = -\mathbf{v}(+\Gamma), \quad E(-\Gamma) = E(+\Gamma) \tag{6}$$

Dimensional check: like-dimensional equalities.

E7: local monodromy spectrum

$$\mu_j(K) = \exp(\lambda_j(J_K)/||J_K||_F) \tag{7}$$

Dimensional check: dimensionless exponent.

E8: three-core additive field interaction

$$E_{\text{cross}} = \langle |\sum_i \mathbf{v}_i|^2 \rangle - \sum_i \langle |\mathbf{v}_i|^2 \rangle \tag{8}$$

Dimensional check: same normalized quadratic field measure in candidate and control.

E9: independent closure guard

$$R = Q_* - T_* - \tfrac{1}{2} Y_* \tag{9}$$

Dimensional check: all normalized; equation is not evaluated if T_* or Y_* is absent.

Registered code-to-mathematics steps

1. **S1** [G2; equations E1]: P00/P01 verify E011 provenance and construct equal-method canonical braid probes.
2. **S2** [G3; equations E1, E2, E3]: P02 computes finite-core reduced Jacobians over resolution/core controls and performs exact-label-permutation family separation.
3. **S3** [G3; equations E4, E5]: P03 computes integer-like periods and target-free signed dynamic-response cancellation rankings.
4. **S4** [G3; equations E6, E7]: P04 tests knot-specific monodromy fingerprints and Gamma-reversal operator covariance.
5. **S5** [G3; equations E6]: P05 independently checks odd velocity/even quadratic energy under Gamma reversal.
6. **S6** [G8; equations E4, E8]: P06 tests the three-ring, three-parallel-+z-core candidate against a matched identity-braid unlink control.
7. **S7** [G8; equations E9]: P08 refuses closure without independently derived doublet and second-U1 operators.
8. **S8** [G8; equations E6]: P07 retains a separate analytic parity-odd background-selector diagnostic; it cannot close a physical mechanism without independently supplied dynamics-ready background evidence.
9. **S9** [G8; equations E2, E4, E5, E7, E8, E9]: P09 aggregates all operator-sector statuses without allowing isolated PASS outcomes to imply integrated closure.

7 Algorithm and phase order

P00 source intake; P01 dynamic-probe qualification; P02 anonymous family dynamics; P03 period and signed external-core response; P04 dynamic monodromy; P05 circulation-conjugation covariance; P06 three-ring/three-independent-parallel-core container; P07 inherited parity-odd background selector; P08 independent U1/doublet closure guard; P09 integrated Rosetta. Execution is sequential but scientific dependencies are nonblocking.

8 Sources and provenance

The embedded E011 archive is SHA-256 pinned. Canonical probe braid words and generator rules are frozen in the implementation commitment. They are mathematical representatives used to test operators, not an independent replication of the upstream E011 geometry providers.

9 Gate semantics

G2 qualifies the source plus dynamic-probe foundation. G3 asks whether at least one dynamic operator lane passes. G8 is deliberately hard and closes only if all registered required mechanism sectors close independently. G9 verifies semantic reveal integrity.

10 Numerics and error controls

Finite-core Biot–Savart uses Python/NumPy FP64. P02 spans multiple curve resolutions and finite-core radii. P03 spans multiple period-probe radii/resolutions. P04 requires $J(-\Gamma) \approx -J(+\Gamma)$ and knot-to-knot monodromy separation larger than numerical fingerprint spread. P06 tests pairwise linking, three own-core periods and a recurrent dynamic cross-energy contrast against an identity-braid unlink control.

11 Backend authority

Python/NumPy FP64 is authoritative in v0.2.0. No C++ or GPU lane is promoted.

12 Blindness and reveal

Blind outputs use HMAC anonymous object IDs. Semantic family roles, target pair/composite hypotheses and conjugate-state interpretation remain private until commitment-verified reveal.

13 Statistics and multiplicity

The family test uses leave-one-out nearest-centroid classification on two frozen dynamic features and exact enumeration of all 3-versus-4 label assignments. The period/response phase ranks the complete registered pair search; the best pair is not treated as a population-significance claim.

14 Automatic results

Manifest note: the framework-generated "Output manifest SHA-256" below records the pre-report snapshot used at render time. The authoritative final archive state is the final OUTPUT_MANIFEST.json written during packaging.

Automatic run provenance

Field	Recorded value
Protocol bundle SHA-256	e9645f2e095a8c4edc98b18eac279b444f53f1ec8cbc142941664de5d634d363
Framework schema	SST-FALSIFIER-PROTOCOL-2
Python	3.13.9 (tags/v3.13.9:8183fa5, Oct 14 2025, 14:09:13) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0
Output manifest SHA-256	32f9ded4d8410b61c68ece8ee57aa88e0626df823609ab94f38a582c6336f981

Gate ledger

Gate	Status	Question	Reason
G0	PASS	Are frozen protocol, vendored framework and implementation byte-consistent?	Framework v1.0.6 and the A059 v0.2.0 implementation commitment are byte-consistent.
G1	PASS	Is embedded E011 v0.3.0 present and SHA-qualified?	Embedded E011 v0.3.0 STATIC_READY archive must match its preregistered SHA-256 exactly.
G2	PASS	Do source intake and self-contained canonical dynamic-probe qualification pass?	Source intake and self-contained dynamic probe qualification both PASS.
G3	PASS	Does at least one preregistered dynamic operator lane pass its controls?	At least one target-free dynamic operator lane PASSes; non-PASS sibling lanes remain preserved.
G4	NOT_APPLICABLE	Is there independent held-out physical confirmation?	No independent held-out physical experiment is claimed in v0.2.0.
G5	NOT_APPLICABLE	Is native C++ certification enabled?	Python/NumPy FP64 is authoritative in v0.2.0; no native C++ certification lane is promoted.
G6	NOT_APPLICABLE	Is GPU screening enabled and parity-qualified?	GPU screening is disabled in v0.2.0.
G7	NOT_APPLICABLE	Is there independent source-family dynamics replication?	E011 provider multiplicity and canonical braid probes are not independent physical replication.
G8	UNRESOLVED	Does the integrated dynamic/operator Rosetta close every required sector?	Integrated Rosetta cannot close while required sectors remain UNRESOLVED/DEFERRED; explicit FAILs remain preserved.

G9	DEFERRED	After blind finalization, do reveal commitments verify without changing blind results?	Semantic reveal is separate and commitment-verified.

Backend manifest


```
{
  "authority": "PYTHON_NUMPY_FP64_DYNAMIC_PROBES",
  "framework_version": "1.0.6",
  "implementation": {
    "file_count": 20,
    "manifest_sha256": "7b65a6422393655fd9b169ea1b3626fb93f096ff52fbb69b1f4d11d590742f9b",
    "mismatches": [],
    "pass": true
  },
  "note": "A059 v0.2.0 promotes self-contained finite-core dynamic operators on canonical closed-braid
  ↪  probes; E011 remains STATIC_READY provenance rather than independent dynamic evidence.",
  "schema": "A059-BACKEND-MANIFEST-2"
}
```


Source manifest


```
{
  "manifest_sha256": "e7c011dc986ddcfe7aa3213d135755dfd0c54424c8ccf92563c09542fd84fce1",
  "schema": "SST-SOURCE-MANIFEST-2",
  "sources": [
    {
      "catalog_id": "E011",
      "evidence_class": "simulation",
      "exists": true,
      "independence_family": "E011_SKLSA_STATIC_READY_V030",
      "path":
      ↪  "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A059_SST_Quantum_Number_Roset
      ↪  ta_Mega_Falsifier\\A059-v0.2.0\\inputs\\E011_SKLSA_Static_Ready_Seed_Atlas_v0.3.0-outputs.zip",
      "provenance_family": "E010_E011_STATIC_GEOMETRY_CHAIN",
      "registry_hint": {
        "catalog_id": "E011",
        "hits": []
      },
      "role": "primary static geometry/provenance atlas",
      "sha256": "97cd1acbf23c102e4564f2d74fd329a2d03f32a29b8bd376ffa8b806c14246cc",
      "source_id": "E011_STATIC_ATLAS",
      "version": "v0.3.0"
    }
  ],
  "workbench_root": "C:\\workspace\\projects\\SST-Workbench"
}
```


15 Allowed interpretation

A PASS certifies only the named finite-model operator/covariance on the frozen canonical probes. A dynamic family separation is not a particle-family identification; a period/cross response is not yet a calibrated physical charge; reduced monodromy is not a quantum-spin proof; and a three-core container result is not by itself a composite-particle identification.

16 Reproducibility

From the A059-v0.2.0 folder run `run_all.cmd` (blind default), `run_all.cmd REVEAL`, `run_all.cmd ALL`, or `run_all.cmd VERIFY`. Each phase writes a separate folder under `PHASES/`; framework gates are mirrored under `GATES/`.